

No credit will be given unless you show your work. Unless stated otherwise, each question is worth 5 pts.

Solve the following equations.

1. $\frac{7x}{8} - 2 = 3 - \frac{15x}{16}$

$$\begin{aligned} 16\left(\frac{7x}{8} - 2\right) &= 16\left(3 - \frac{15x}{16}\right) \\ 14x - 32 &= 48 - 15x \\ 29x &= 80 \\ x &= \frac{80}{29} \end{aligned}$$

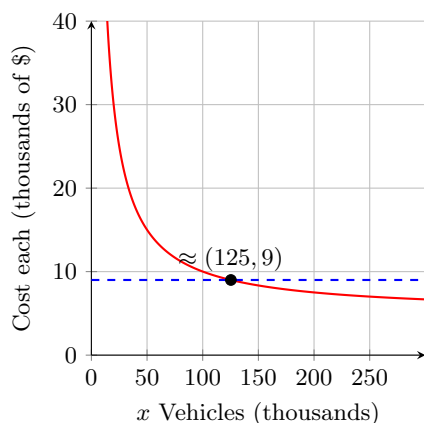
2. $4|x + 1| - 7 = 6$

$$\begin{aligned} 4|x + 1| &= 13 \\ |x + 1| &= \frac{13}{4} \\ x + 1 &= \pm \frac{13}{4} \\ x &= \pm \frac{13}{4} - 1 = -\frac{17}{4}, \frac{9}{4} \end{aligned}$$

Check!

Use the provide graph to answer the question.

3. The manufacturing cost per vehicle for production of x vehicles is shown in the graph. Use the graph to estimate the number of vehicles that must be produced so that the cost per vehicle is \$9,000.



The graph has a height (y -coordinate) of 9 at about $x = 125$. So 125,000 vehicles should be produced.

Solve the following equation for the provided variable.

4. $G = Q + (M - 7)P$ for M .

$$\begin{aligned} G &= Q + MP - 7P \\ G - Q + 7P &= MP \\ \frac{G - Q + 7P}{P} &= M \end{aligned}$$

Solve the following word problems

5. Joe bought a pair of high heels for \$322.5 including tax. If the sales tax is 7.5% then what was the cost of the high heels before tax?

$$\begin{aligned} 322.5 &= (1 + 0.075)x \\ \frac{322.5}{1.075} &= x = \$300.0 \end{aligned}$$

6. Pavlos, the Athenian executioner, needs to make some hemlock poison. How many gallons of a 80% hemlock solution must be mixed with 10 gallons of 35% hemlock to get a mixture that is 65% hemlock.

$$\begin{aligned} (.80)x + (.35)10 &= (.65)(x + 10) \\ .8x - .65x &= 6.5 - 3.5 \\ .15x &= 3 \\ x &= 3/.15 = 20 \text{ gallons} \end{aligned}$$

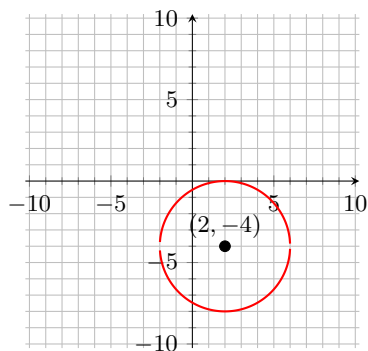
7. John can clean the house in 45 minutes. James can clean the same house in 15 minutes. How long would it take them to clean the house if they work together?

$W = rt$. So 1 job is equal to the sum of the two rates time t ,

$$\begin{aligned} \left(\frac{1}{45} + \frac{1}{15}\right)t &= 1 \\ t &= \frac{45}{4} = 11.25 \text{ minutes} \end{aligned}$$

Perform the indicated operation(s).

8. Find the center and the radius of the circle. Then graph the circle $(x - 2)^2 + (y + 4)^2 = 16$.



The center is $(2, 4)$ and the radius is $\sqrt{16} = 4$.

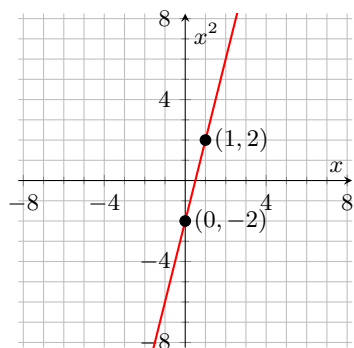
9. Write the standard equation for the circle centered at $(9, 4)$ and passing through the origin.

The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$ where r = radius and the center is (h, k) . The distance from the center to the origin, $(0, 0)$, is $\sqrt{(0 - 9)^2 + (0 - 4)^2} = \sqrt{25} = 5$. So $r = 5$ and $(x - 9)^2 + (y - 4)^2 = 25$

10. Find the slope, if it exists, of the line passing through the points $(5, 0)$ and $(-2, 1)$.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0}{-2 - 5} = -\frac{1}{7}$$

11. Write an equation for the line shown in the graph.



$$m = \frac{2 - (-2)}{1 - 0} = 4 \text{ and } y - y_1 = m(x - x_1) \text{ gives,}$$

$$\begin{aligned} \Rightarrow y + 2 &= 4(x - 0) \\ y &= 4x - 2 \end{aligned}$$

12. Find an equation for the line which passes through the point $(0, 1)$ and is perpendicular to $y = 3x + 4$.

$$m = -\frac{1}{3} \text{ and } y - 1 = -\frac{1}{3}(x - 0) \Rightarrow y = -\frac{1}{3}x + 1$$

Perform the indicated operations and write your answers in the form $a + bi$.

13. $(2 - i)(4 + 8i)$
 $= 4 - 4i + 10i - 8i^2 = 4 + 6i - 8(-1) = 12 + 6i$

14. $\frac{4}{5+i}$
 $\left(\frac{4}{5+i}\right)\left(\frac{5-i}{5-i}\right) = \frac{20-4i}{25-i^2} = \frac{20-4i}{24} = \frac{20}{24} - \frac{4i}{24} = \frac{5}{6} - \frac{1}{6}i$

15. [2 pts] i^5
 $i^5 = i^2 i^2 i = (-1)(-1)i = i$ note that $i^4 = 1$

16. [3 pts] i^{24}
 $i^{24} = (i^4)^6 = (1)^6 = 1$

Use the square root property to find all real or imaginary solutions.

17. $x^2 - 17 = 0$

$$\begin{aligned} x^2 &= 17 \\ \sqrt{x^2} &= \pm\sqrt{17} \\ x &= \pm\sqrt{17} \end{aligned}$$

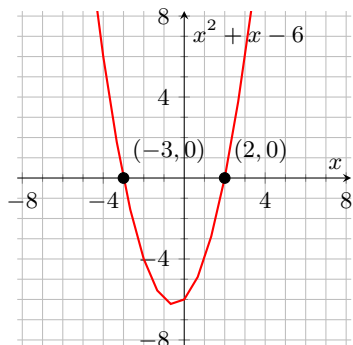
Solve for x .

18. $2x^2 - 36 = 6x$

$$\begin{aligned} 2x^2 - 6x - 36 &= 0 \\ 2(x^2 - 3x - 18) &= 0 \\ 2(x-6)(x+3) &= 0 \\ x &= 6, -3 \end{aligned}$$

Solve for x .

19. $x^2 + x - 6 = 0$



From the graph,
 $x = -3$ and $x = 2$